

Mt. Washington MANET Field Trial

Trial 1 — 2026-05-23

Ethan Doherty CS Graduate Directed Study, Summer 2026

Ammonoosuc Ravine Trail (ascent) · Jewell Trail (descent) · 915 MHz LoRa · Meshtastic mesh

Overview

This report documents the first empirical field trial of a Meshtastic-based LoRa mesh radio network on Mt. Washington, NH. The intended objective was to characterize LoRa signal propagation along a roughly 10-hour ascent/descent route and screen the feasibility of a low-cost fixed-relay infrastructure for autonomous wilderness emergency communications. The collector recorded only 6 h 9 m of that route, and the ambient packets lack the controlled endpoints and direct-hop labels needed to characterize a propagation link.

A Raspberry Pi running a custom telemetry collector served as the mesh HEAD node, logging all received RF packets (RSSI, SNR, mesh node ID, GPS coordinates, UTC timestamp) to an append-only JSONL stream. GPS ground truth was independently provided by a Garmin wearable GPS unit carried simultaneously.

Timeline of Events

Time (EDT)	Event
08:06	Departed trailhead (Base Rd, $\approx$ 764 m). Garmin GPS active. Collector not yet online.
09:36	Collector came online briefly, then entered reconnect loop.
09:36 – 12:24	Collector offline (2 h 48 m). Meshtastic Python library lost serial connection and looped silently. No RF packets logged. GPS track unaffected.
12:24	Collector reconnected. Active RF telemetry resumed.
14:23	Summit reached — 1917 m. Peak node density: $\sim$ 22 unique nodes in rolling 30-min window.
14:23 – 18:33	Descent via Jewell Trail. Signal quality tracked in 5-min RSSI windows.
18:33	Collector stopped. The hike-map input contains 686 packet events across the afternoon active window; the separately generated mesh catalog reports 764 RF observations from the earlier pre-hike soak. The two totals have been reconciled (<code>scripts/reconcile_trial1_counts.py</code>): they are distinct time windows on the same decoded receiver log, not one population, so neither is a single “nodes heard” total.
18:40	Returned to trailhead.

Observed RF Performance

Metric	Value
Packet count	686 hike-map events; 764 RF observations in alog
Active collection window	12:24 – 18:33 EDT (6 h 9 m)
Unique-node count	50 decoded IDs in map input; 41 RF source IDs catalog
Nodes with GPS position	24 (per mesh catalog; from received POSI packets)
Peak concurrent nodes (30-min window)	~22 (at summit)
Best RSSI observed	–38 dBm
Analyzed 5-min descent windows with ≥ 1 packet from any source	100%

The count differences above are reconciled, not sampling uncertainty: they come from two different time windows on the same decoded receiver log. The 686 records / 50 IDs are the afternoon hike window ($\geq 16:24$ UTC, all packet types); the 764 observations / 41 IDs are the pre-hike soak snapshot (before 02:34 UTC, rssi-bearing packets). The reconciliation script (`scripts/reconcile_trial1_counts.py`) reproduces both the 686/50 numbers and the exact 41-node catalog ID set from that single log, and finds 18 IDs in both windows, 32 hike-only, and 23 catalog-only. In every analyzed 5-minute window of the descent interval (summit at 14:23 until the collector stopped at 18:33), the HEAD node demodulated at least one Meshtastic packet from *some* source — any of ~50 ambient mesh nodes at unknown distances, potentially relayed. This is evidence of ambient 915 MHz mesh activity reaching forested sub-alpine terrain below 1200 m; it is *not* evidence of bidirectional coverage, of reachability to any particular gateway, or of any specific link’s viability. Those claims require the controlled-link methodology planned for Trial 2 (Section 8).

Issues Encountered

Issue 1 — Collector outage (2 h 48 m)

The Meshtastic Python library (`meshtastic.serial_interface`) lost its serial connection to the LoRa radio and entered a silent reconnect loop. No exception was raised; the gap was discovered post-hike via timestamp analysis of the JSONL stream. This is a software/serial-path outage in this record; the available evidence does not isolate the library as the sole cause or rule out a hardware/USB contribution. The gap covered a substantial part of the Ammonoosuc Ravine ascent, so that interval has GPS ground truth but no RF observations.

Issue 2 — No GPS pairing before departure

The operator’s phone GPS was not paired with the Meshtastic network before leaving the parking lot. The HEAD node’s own position was therefore not broadcast into the mesh during the trial. The Garmin track later supplies receiver position, but it cannot supply the original transmitter position, direct-hop identity, or synchronized link geometry.

Issue 3 — No hop-count ground truth

The collector version deployed during Trial 1 predated the addition of `hop_limit` and `hop_start` fields. It is therefore not possible to determine whether any given packet was a direct link or a mesh relay hop. Trial 2 will address this with a dual-radio setup (see Section 8).

Propagation Model

Log-Distance Path Loss

The standard log-distance path loss model used across LoRa mountain SAR research [1, 3] is:

$$PL(d) = PL(d_0) + 10n \log_{10}\left(\frac{d}{d_0}\right) + X_\sigma \quad (1)$$

where d_0 is a reference distance, n is the path loss exponent, and X_σ is a zero-mean Gaussian random variable with standard deviation σ (dB) representing log-normal shadowing. An alternative floating-intercept form [3] is:

$$PL(d) = \alpha + 10\beta \log_{10}(d) + X_\sigma \quad (2)$$

where α and β are fitted from measurements. The two forms are related by $\beta = n$ and $\alpha = PL(d_0) - 10\beta \log_{10}(d_0)$; anchoring the reference distance in free space gives $PL(d_0) = 20 \log_{10}(4\pi d_0/\lambda)$.

Bianco et al. [1] measured $n \approx 2.0$ – 3.5 in mountainous terrain with $\sigma \approx 6$ – 8 dB, and found that standard propagation models significantly underpredict path loss at 915 MHz relative to measurements, with α values deviating by up to 10 dB over the full LoRa radio range.

Free-Space Path Loss Baseline

The free-space path loss ($n = 2$, $X_\sigma = 0$) at 915 MHz serves as the lower bound on path loss in this model:

$$FSPL(d) = 32.44 + 20 \log_{10}(d_{\text{km}}) + 20 \log_{10}(f_{\text{MHz}}) \quad (3)$$

At 915 MHz this evaluates to $FSPL(d) = 91.65 + 20 \log_{10}(d_{\text{km}})$ dB.

Terrain Loss Augmentation

Because standard propagation models are insufficient for heterogeneous mountain terrain [3], an empirical terrain-class loss term L_T is added to Eq. (3):

$$PL_{\text{aug}}(d, z) = FSPL(d) + L_T(z) \quad (4)$$

where the terrain class is determined by elevation z :

$$L_T(z) = \begin{cases} 3 \text{ dB} & z \geq 1500 \text{ m} \quad (\text{alpine, open}) \\ 15 \text{ dB} & 1200 \leq z < 1500 \text{ m} \quad (\text{sub-alpine transition}) \\ 25 \text{ dB} & z < 1200 \text{ m} \quad (\text{forested}) \end{cases} \quad (5)$$

The forested value of 25 dB is consistent with the body of literature on vegetation attenuation at 900 MHz-band frequencies, where canopy excess loss of 20–30 dB is commonly reported for dense mixed forest at ranges beyond 1 km [3].

Received Signal Power and Link Budget

The predicted received signal power is:

$$P_r(d, z) = P_t + G_t + G_r - PL_{\text{aug}}(d, z) \quad (6)$$

where $P_t = 22$ dBm (Heltec LoRa32 V3 maximum TX power), and $G_t = G_r = 2.15$ dBi (dipole antenna gains). The total link budget is:

$$\mathcal{L} = P_t + G_t + G_r - P_{\text{sens}} = 22 + 2(2.15) - (-131) = 157.3 \text{ dB} \quad (7)$$

The receiver sensitivity $P_{\text{sens}} \approx -131$ dBm corresponds to SF11, BW = 250 kHz, CR = 4/5 — the Meshtastic *LongFast* modem preset used during this trial (Meshtastic documents a 153 dB link budget at 22 dBm TX with 0 dBi antennas, implying $P_{\text{sens}} \approx -131$ dBm, consistent with the Semtech SX1262 datasheet). Note that SF12/BW 125 kHz is the *deprecated LongSlow* preset, not LongFast.

LoRa Physical Layer Parameters

The data rate of a LoRa symbol is determined by the spreading factor and bandwidth [4]:

$$R_s = \frac{BW}{2^{SF}} \quad (8)$$

$$DR = SF \cdot \frac{BW}{2^{SF}} \cdot CR \quad (9)$$

Each unit increase in SF provides approximately 2–3 dB improvement in SNR sensitivity at the cost of a halved data rate [4]. The LongFast preset (SF11, BW = 250 kHz, CR = 4/5) yields a data rate of ≈ 1.07 kbps and a sensitivity floor of ≈ -131 dBm; the deprecated SF12/BW 125 kHz LongSlow preset would yield ≈ 0.18 kbps at ≈ -137 dBm.

Packet Delivery Ratio

Bianco et al. [1] reported that the path loss at the 50% packet delivery ratio (PDR) threshold is approximately 135 dB in mountainous canyon terrain. The measured range at $\geq 50\%$ PDR was 300 m in worst-case canyon geometry — more than five times the operating range of standard ARVA avalanche beacons under the same conditions. In open alpine terrain, ranges extended to several kilometres.

The effective signal power for LoRa when $SNR < 0$ dB is given by [3]:

$$ESP = RSS + SNR - 10 \log_{10}(1 + 10^{0.1 \cdot SNR}) \quad (10)$$

Historical Relay Screen (Superseded by Terrain-Profile Analysis)

Node Placement

Three candidate nodes were initially selected at GPS-derived treeline locations using an elevation-class/FSPL heuristic. The original assertion that each location had simultaneous line-of-sight uphill and downhill was not based on a terrain-profile test and is false for the summit-facing links; Section 7 supersedes it.

Node	Role	Latitude	Longitude	Elev.
Ammo Relay	Treeline relay	44.26616°N	71.32348°W	1201 m
Jewell Relay	Treeline relay	44.28376°N	71.33583°W	1199 m
Trailhead Gateway	Starlink uplink	44.26700°N	71.36083°W	764 m

Historical FSPL-Screen Predictions

Applying Eq. (6) with the terrain augmentation of Eq. (5):

Link	Distance	Terrain	Predicted RSSI	Margin above P_{sens}
Ammo Relay → Summit	1.68 km	Alpine	−72.9 dBm	58.1 dB
Jewell Relay → Summit	2.98 km	Alpine	−77.9 dBm	53.1 dB
Ammo Relay → Gateway	2.98 km	Forest	−99.8 dBm	31.2 dB
Jewell Relay → Gateway	2.73 km	Forest	−99.1 dBm	31.9 dB

Within this historical non-geometric screen, all four links clear the assumed receiver sensitivity floor (−131 dBm) by more than 31 dB. The earlier analysis subtracted 3σ using a shadowing parameter reported in another Alpine study [1]; that does not establish a 99.87% reliability level for these sites. The sites, measurement protocol, residual distribution, canopy, and terrain blockage were not shown to match, and the subsequent terrain-profile results reverse the summit conclusion.

Model limitation. Eq. (6) with the elevation-class loss of Eq. (5) contains no terrain-profile geometry: no line-of-sight test, no Fresnel-zone clearance, and no knife-edge diffraction. These link predictions are therefore *screening estimates only*. Section 7 re-evaluates every proposed link with the Longley–Rice ITM model over real USGS 3DEP terrain — and **reverses the conclusions of this table for the summit-facing links**.

Maximum Planning Range

Range is computed against an explicit *planning threshold* $P_{\text{thr}} = -100$ dBm rather than the absolute sensitivity floor. The ≈ 31 dB standoff from $P_{\text{sens}} \approx -131$ dBm provides a fade margin covering $> 3\sigma$ log-normal shadowing ($\sigma \approx 7$ dB [1]) plus interference and hardware aging. Setting $P_r = P_{\text{thr}}$ in Eq. (6), with $PL_{\text{max}} = P_t + G_t + G_r - P_{\text{thr}} = 126.3$ dB:

$$d_{\text{max}} = 10^{\frac{PL_{\text{max}} - L_T - 91.65}{20}} \text{ km} \quad (11)$$

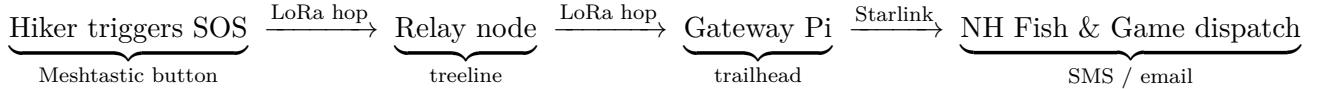
$$d_{\text{max}}^{\text{alpine}} = 10^{(126.3 - 3 - 91.65)/20} = 38.2 \text{ km}$$

$$d_{\text{max}}^{\text{forest}} = 10^{(126.3 - 25 - 91.65)/20} = 3.0 \text{ km}$$

(Solving instead for $P_r = P_{\text{sens}}$ gives FSPL-only ranges exceeding 1,000 km — a physically meaningless extrapolation, since free space with a flat terrain-class constant ignores diffraction and terrain blockage entirely; this is why a planning threshold is used.) This historical model produced the statement that the 2 h 48 m collector gap “would have been covered.” That is not a defensible coverage claim: it ignores the blocking terrain that causes the ITM result in Section 7. The later model screen is itself unvalidated and does not replace field PDR measurement.

Automation Chain

The proposed concept would require the following chain:



Target end-to-end latency: <30 seconds. Trial 1 did not implement or test this chain, gateway reachability, dispatch integration, acknowledgments, authorization, or agency workflow, so the target is a requirement rather than a result. The historical rough BOM was $\approx$ \$690 (two relay nodes at $\sim$ \$120 each, gateway Pi + radio $\sim$ \$150, Starlink Mini $\sim$ \$300); prices, power hardware, mounting, weatherization, service, permits, maintenance, and any existing backhaul must be quoted before this is presented as a deployment cost.

Terrain-Profile Verification (ITM/Longley–Rice)

Every proposed link was re-evaluated with the Longley–Rice Irregular Terrain Model (ITM v7.0, point-to-point mode) via the `itmlogic` reference implementation [5], using terrain profiles sampled at 15 m from a $\sim$ 7 m/px USGS 3DEP elevation export. Parameters: 915 MHz, vertical polarization, continental temperate climate, $N_s = 301$, $\epsilon_r = 15$, $\sigma = 0.005$ S/m; antenna heights 3 m (relay/gateway masts) and 1.5 m (hiker handheld); 50% confidence. Predicted RSSI = $P_t + G_t + G_r - L_b(q)$ at reliability quantiles q . Pipeline: `scripts/dem_3dep.py` + `scripts/itm_relay_links.py`; artifacts in `artifacts/itm/`.

Proposed links over real terrain

Link	Dist	RSSI q_{50}	RSSI q_{90}	F_1 clear.	Verdict
Ammo Relay $\rightarrow$ Summit	1.68 km	−132.2	−132.3	−5.34	below sensitivity
Jewell Relay $\rightarrow$ Summit	2.98 km	−118.7	−118.7	−1.90	marginal (no margin)
Ammo Relay $\rightarrow$ Gateway	2.98 km	−76.6	−76.6	+0.89	strong
Jewell Relay $\rightarrow$ Gateway	2.73 km	−74.1	−74.1	+1.19	strong
Ammo $\leftrightarrow$ Jewell Relay	2.19 km	−130.1	−130.2	−4.51	below margin

(F_1 clear. = worst first-Fresnel-zone clearance fraction with $4/3$ effective Earth radius; ≥ 0.6 is the usual engineering criterion, negative means the geometric ray is below terrain.)

ITM reverses the screening model. The FSPL + terrain-class table in Section 6.2 predicted the summit links as the strongest (−72.9 dBm) and the forested gateway links as the weakest (−99.8 dBm). Over real terrain the opposite holds: the valley-crossing gateway links are clear-LOS and strong, while the treeline-to-summit chord is blocked by the *convexity of the summit cone itself* — terrain rises up to 53 m above the direct ray at $\sim$ 1.2 km from the Ammo relay. Mitigations tested and rejected: a 10 m relay mast (−131.9 dBm; the obstruction is mid-path, not local), relocating

to the Lakes of the Clouds shelf at 1,529 m (−121.5 dBm to summit, still diffracted), and a 10 m summit-tower node (−113.2 dBm to the gateway — blocked in the other direction). At 915 MHz with practical antenna heights, *no single-hop link from below clears the upper cone*.

Collector-gap segment, re-examined

Sampling the Garmin track across the 2 h 48 m collector gap (09:36–12:24 EDT, 84 points at 120 s) and taking the best of the two proposed relays per point (ITM q_{90}): **100%** of points are predicted above raw sensitivity (−131 dBm), but only **45.2%** clear the −100 dBm planning threshold, and 33% have geometric line-of-sight; the worst point is −121 dBm at 3.07 km. The Section 6.3 claim that the gap “would have been covered” must therefore be downgraded: coverage of the Ammonoosuc Ravine ascent is *marginal* — physically plausible at the assumed sensitivity but with zero fade margin and unmodeled canopy loss below treeline. It cannot be claimed as robust.

This disagreement is itself a measurable hypothesis. The screening model and ITM differ by up to ~60 dB on the summit links. Trial 2 will place a beacon node at the proposed Ammo relay coordinates and measure PDR while ascending the ravine (Section 8); the observed PDR-vs-position profile discriminates directly between the two models and grounds any deployment proposal in measurement rather than prediction.

Airtime and SOS-chain capacity

LongFast (SF11, BW 250 kHz, CR 4/5, 16-symbol preamble): $T_{\text{sym}} = 2^{11}/250 \text{ kHz} = 8.192 \text{ ms}$. Computed airtimes (Semtech AN1200.13 formulation; `scripts/lora_airtime.py`): position packet (~40 B) 559 ms, SOS text (~64 B) 723 ms. A 3-hop SOS chain is ~2.2 s of serial airtime; with per-hop CSMA backoff under light load, end-to-end delivery remains seconds — the <30 s target has order-of-magnitude margin *when the channel is lightly loaded*. The caveat is load: one position beacon per node per 60 s costs 0.93% airtime per node, and the ~22 concurrent nodes observed at the summit imply ~20% raw channel utilization before mesh rebroadcast amplification. Congestion management (beacon intervals, hop limits, ideally a dedicated channel for infrastructure traffic) must be part of any deployment plan.

Plans Going Forward

Trial 2 — Hop-count ground truth

Two Heltec V3 radios may be connected to the Pi simultaneously for redundancy:

- **Radio A:** restricted to `maxHops=1`
- **Radio B:** unrestricted (`maxHops=3`)

The two-radio comparison does *not* by itself establish hop truth: independent packet loss can make a direct packet appear on only one receiver, and a hop-limit setting is not a direct-link label. Calibration eligibility is determined per packet from recorded `hop_start/hop_limit` metadata with `hops_away == 0`, a dedicated trial channel, monotonic transmitter sequence numbers, and surveyed endpoints. Receiver logs must also record which physical radio produced each observation.

Trial 2 — Pre-departure checklist

1. Pair phone GPS before leaving parking lot; verify `lat/lon` updating in JSONL
2. Place static relay nodes at known waypoints with recorded coordinates

3. Run `head_readiness_report.py` from parking lot before departure
4. Confirm no restart timers active on the Pi

Infrastructure proposal

Engage NH Fish & Game and Mt. Washington State Park to evaluate a pilot deployment of relay nodes at the treeline crossings identified above.

Research outreach

Contact authors of the relevant Alpine LoRa studies only after verifying their current affiliations and contact details. Their measurements [1, 2] provide methodological comparison data; they do not calibrate or validate this project’s terrain model.

Calibration Methodology and Trial 1 Findings

The AIRMap calibration pipeline was implemented and executed against the Trial 1 telemetry. The pipeline computes a free-space path loss baseline (Eq. 3), joins predicted and observed signal metrics by timestamp, and fits the floating-intercept model of Eq. 2 by ordinary least squares on *calibration-eligible* rows only (GPS on both ends, verified direct link, plausible received power, fresh position fixes), using effective signal power (ESP, Eq. 10) as the observable where SNR is available. Evaluation uses contiguous-time blocked cross-validation; parameter uncertainty is estimated with a moving-block bootstrap. A constant mean-bias correction is retained only as a comparison model. The Garmin GPS track was used as HEAD position ground truth.

Why Trial 1 did not produce valid calibration data

Post-hoc analysis of the calibration dataset identified three structural measurement gaps that rendered the output numerically unreliable:

1. **No hop-count field.** The collector version deployed in Trial 1 did not capture `hop_limit` or `hop_start`. Meshtastic logs the RSSI of the *last hop*, not the original transmitter. Without hop count it is impossible to filter to direct links, making it impossible to associate an observed RSSI with a known source-to-receiver distance.
2. **Adjacent device forwarding.** A second Heltec radio (!db51af80) was carried co-located with the HEAD node. This device forwarded packets from distant Meshtastic users across NH/VT/ME. The HEAD logged these forwarded packets with the original sender’s GPS coordinates (55–183 km away) but the adjacent device’s RSSI (−1 to −18 dBm). The resulting distance–RSSI pairs are physically invalid and contaminate the calibration dataset.
3. **No independently placed static node.** The trial did not include a second radio at a fixed, independently measured GPS location. Without a controlled source-to-receiver link, there is no ground-truth distance–RSSI pair against which to calibrate.

What Trial 2 will produce

Trial 2 deploys `meshnode1` as a controlled transmitter at a surveyed site while the HEAD follows a GPS-tracked route. Only observations that pass direct-hop, endpoint-position, time-sync, sequence-denominator, and plausibility gates enter calibration. Other packets remain in the raw dataset and are labelled ineligible. The feasible sampling amendment and exact opportunity accounting are recorded in `docs/trial2-preregistration.md`.

References

- [1] G. M. Bianco, R. Giuliano, G. Marrocco, F. Mazzenga, and A. Mejia-Aguilar, “LoRa System for Search and Rescue: Path-Loss Models and Procedures in Mountain Scenarios,” *IEEE Internet of Things Journal*, vol. 8, no. 3, pp. 1985–1999, 2021. DOI: 10.1109/JIOT.2020.3017044
- [2] G. M. Bianco, A. Mejia-Aguilar, and G. Marrocco, “Performance Evaluation of LoRa LPWAN Technology for Mountain Search and Rescue,” in *Proc. 5th International Conference on Smart and Sustainable Technologies (SpliTech)*, 2020, pp. 1–4. DOI: 10.23919/SpliTech49282.2020.9243817
- [3] J. A. Azevedo and F. Mendonça, “A Critical Review of the Propagation Models Employed in LoRa Systems,” *Sensors*, vol. 24, no. 12, p. 3877, 2024. DOI: 10.3390/s24123877
- [4] Q. Guo, F. Yang, and J. Wei, “Experimental Evaluation of the Packet Reception Performance of LoRa,” *Sensors*, vol. 21, no. 4, p. 1071, 2021. DOI: 10.3390/s21041071
- [5] E. J. Oughton, T. Russell, J. Johnson, C. Yardim, and J. Kusuma, “itmlogic: The Irregular Terrain Model by Longley and Rice,” *Journal of Open Source Software*, vol. 5, no. 51, p. 2266, 2020. DOI: 10.21105/joss.02266